

HerMin Textile: Competing in the fashion market using Google and TensorFlow

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ABOUT HERMIN TEXTILE

HerMin Textile uses TensorFlow and Google Cloud to build a recommendation learning system and an application to produce designs faster to customers, and it enables designer collaboration.



About HerMin Textile

Founded in 1976 and headquartered in Taiwan, HerMin Textile Co. provides woven fabrics for apparel businesses worldwide.

Industries: Manufacturing

Location: Taiwan

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Google Cloud results

- Enabled the business and its customers to compete more effectively against fast fashion businesses and new fashion designers
- Established a system to enable the business to create more than 10,000 designs in the next two years
- Enabled the business to progress plans to establish a SaaS application to enhance collaboration

Reduced
to bring
design
market



About CloudMile

Founded in 2016, CloudMile is a Taiwan-based B2B cloud service provider that offers tailor-made cloud solutions to businesses that leverage the power of big data analysis and machine learning.

businesses that leverage the power of big

Cloud Datalab (<https://cloud.google.com/datalab/>)

Cloud DNS (<https://cloud.google.com/dns/>)

Cloud Storage (<https://cloud.google.com/storage/>)

Compute Engine
(<https://cloud.google.com/compute/>)

within the
design
community

Founded in 1976 and headquartered in Taiwan, HerMin Textile Co. (<https://www.hermin.com/>) provides woven fabrics for apparel businesses worldwide.

The company's mission is to provide high-quality, sustainable, and environmentally friendly products for prestigious clients such as Ralph Lauren, J. Crew, Filson, and Burberry. HerMin Textile is widening the range of products made available to customers by adding lycra, nylon, and organic cotton to traditional fibres such as cotton, silk, linen, and wool.

Fast fashion, new designer pressure

Ongoing investments in design, research and development, and technology are crucial to the company's success. According to Neil Lee, an NST Capital digital consultant working with HerMin Textile, the business and its clients are coming under increased competitive pressure from fast fashion and new designer brands. In particular, HerMin Textile's approach of storing thousands of sample textile designs in a warehouse that employees had to visit so they could photograph designs for customers was proving costly and inefficient.

Machine learning for core strengths

Following a comprehensive internal review, and with input from Google Cloud Partner [CloudMile](https://www.mile.cloud/gcp/about-us/)

(<https://www.mile.cloud/gcp/about-us/>), HerMin Textile decided to use machine learning to capitalise on its core strengths of textile design and production.

“Having accrued many years of experience and skill, HerMin Textile wanted to preserve its processes and outputs as data that could be analysed, replicated, and built upon,” Lee says. “The business saw an opportunity to use machine learning to identify textile patterns such as blue stripes and improve the process of selecting patterns of value to its customers.”

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—Neil Lee, Consultant to
HerMin Textile

Modernising communications with a mobile app

HerMin Textile wanted the machine learning project to power a mobile application that would modernise communications with and between designers. “A lot of designers use phones and tablets at work, so the business wanted to create a mobile application to enable them to complete tasks on the fly and communicate easily with other designers or their customers,” Lee says. “For example, the application would enable designers to access a database of patterns and, using their mobile phone or tablet, select those that they needed.”

TensorFlow: the best fit

With the assistance of CloudMile, HerMin Textile evaluated cloud-based machine learning products and decided the TensorFlow (<https://www.tensorflow.org/>) open source software library was the best technology on which to build its machine learning system. “The combination of a mature, large developer community and support from Google made TensorFlow the best choice for the business,” Lee says.

The textile business formally selected TensorFlow and Google Cloud (<https://cloud.google.com/>) and commenced the project with CloudMile in October 2017. The machine learning-based textile pattern recognition module and mobile application are expected to be released into production in June or July 2018.

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—Neil Lee, Consultant to
HerMin Textile

HerMin Textile provides labelled images of textile designs to a CloudMile data engineer, who uses [Google Cloud Datalab](https://cloud.google.com/datalab/) (<https://cloud.google.com/datalab/>) to undertake data analysis and develop and hypertune machine learning models. [Google Compute Engine](https://cloud.google.com/compute/) (<https://cloud.google.com/compute/>) provides the virtual infrastructure resources required to run the mobile application and website, while the machine learning models are stored in [Google Cloud Storage](https://cloud.google.com/storage/) (<https://cloud.google.com/storage/>). HerMin Textile also sends the images themselves to Google Cloud Storage, while image search requests and responses are also managed on Google Compute Engine. The

data engineer can also access the images directly from Google Cloud Storage.

Manual warehouse process automated

Rather than locating designs and sending individual digital photographs to customers—a process that may take several days if multiple images or designs are involved—HerMin Textile will be able to share several patterns at once. The ability to meet customers' requirements will increase as the machine learning system will enable the business to select similar or relevant patterns for their use. "The machine learning system and a database of patterns will enable HerMin Textile to create more than 10,000 new patterns in two years," Lee says.

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HerMin Textile

Speed to market increased by 25%

The business estimates the combination of the machine learning system and application will reduce the time needed to bring a new design to market by as much as 25%. “In general, bringing a new clothing design to the market takes about 12 months,” Lee explains. “With these new systems, we expect to reduce this period to about 9 months.”

Building a community of designers

HerMin Textile also expects the mobile application to create a community of designers that can share images and opinions, fostering knowledge and creativity within the industry. Longer term, the business is planning to make the machine learning and mobile application products available to other industry participants as a service from the cloud.